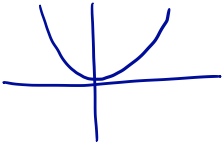
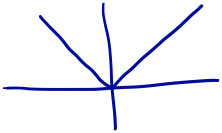
Strong Convexity and Smoothness

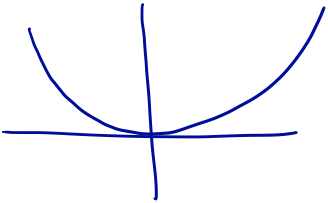
$f(x) = a x^2$ vs $f(x) = |x|$





faster convergence here.

Smoothness and Self-Tuning — 1/3



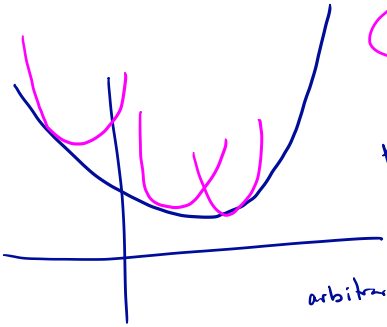
GD:

$$x_{t+1} = x_t - \underbrace{\overset{\text{step size}}{\eta} \nabla f(x_t)}_{\text{update}}$$

$$x_{t+1} = x_t - \underbrace{\overset{\text{fixed}}{\eta} 2a x_t}_{\text{update} \to 0 \text{ as } x_t \to x^*.}$$

$(x^* = 0)$

Smoothness and Self-Tuning — 2/3



$\eta \nabla f(x_t)$

A function will have the self-tuning property that allows us to get arbitrarily close to x^* without needing to have arb. small η: if we can <u>upper bound</u> by a quadratic.

$$f(y) \le f(x) + \nabla f(x)^T (y-x) + \frac{M}{2} \|y-x\|_2^2$$

// Upper bounding f by a quadratic.

Equivalent to ∇f M-Lipschitz: $\|\nabla f(x) - \nabla f(y)\|_2 \le M \cdot \|x-y\|_2$ /

Smoothness and Self-Tuning — 3/3

Self - Tuning

Quad Upper Bound

Lipschitz Gradients

Const. step size : $\eta = 1/M$

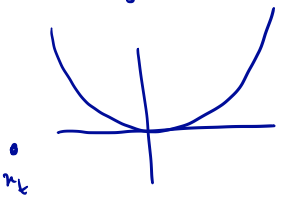
$\frac{1}{\varepsilon}$ Guarantees that GD convergence rate

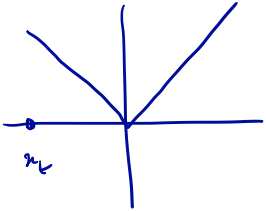
$$O(1/T)$$

iterations.

Strong Convexity — 1/3

$f(x) = ax^2$ vs $f(x) = |x|$





$$x_{t+1} = x_t - 2\eta a \cdot x_t$$

$$x_{t+1} = x_t + \eta \cdot 1$$

magnitude of update same whether x_t is far away or close.

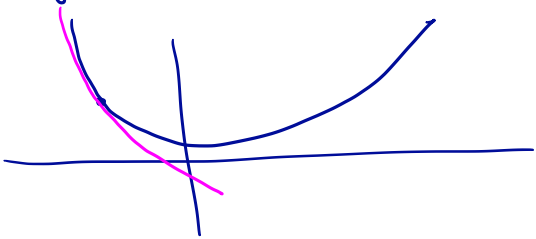
GD makes very fast progress when we are far.

Strong Convexity — 2/3

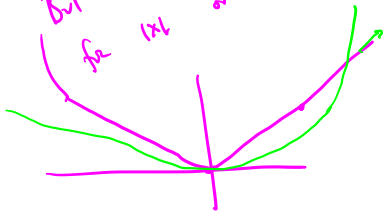
This key property — fast progress when far away — is <u>guaranteed</u>

if we can <u>lower bound</u> f by a quadratic.

$$f(y) \geq f(x) + \nabla f(x)^T (y-x) + \frac{m}{2} \|y-x\|_2^2$$



But: cannot do this for $|x|$ or

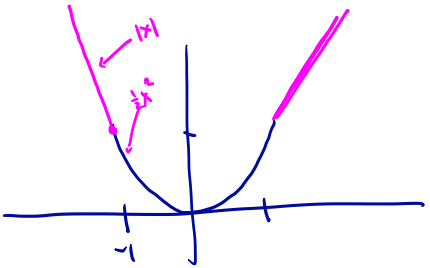


Strong Convexity — 3/3

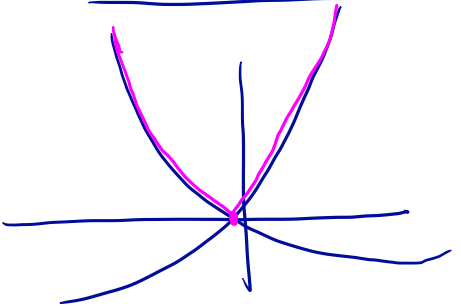
strong convexity : f is lower bounded by a quadratic.

Guarantees big updates $\boxed{\eta \nabla f(x_t)}$ when x_t is far away from x^*.

Smooth but not str convex



Str. convex but not smooth



Strongly Convex & Smooth

If $f : \mathbb{R}^n \to \mathbb{R}$

M – smooth

m – s.c.

GD has linear convergence:

$$f(x_T) - f(x^*) \leq O\left(\left(1 - \frac{m}{M}\right)^T\right)$$